Sanjana Sao

sanjana.sao01@gmail.com | +91 93995 55216 | Noida, Gurgaon | LinkedIn

Professional Summary

AI/ML Engineer with 3.6 years of experience specializing in Computer Vision, NLP, Generative AI, and Large Language Models (LLMs). Experienced in designing, and developing, scalable AI solutions using Python, FastAPI, Docker, and AI services. Proficient in LangChain, Langraph, RAG architectures, AI Agents, MCP (Model Context Protocol), Tool Calling, Vector Databases, and Document Intelligence solutions. Skilled in building end-to-end AI pipelines leveraging transformers, embeddings, multimodal vision models, and agentic workflows to solve real-world business problems. Strong experience in consulting-style delivery, solution design, and production deployment of AI applications.

Technologies

Languages & Libraries: Python, Java, C#, NumPy, Pandas, OpenCV, EDA, scikit-learn, matplotlib, Git

AI/ML & Computer Vision: CNNs, RNNs, LSTMs, Transformers, PyTorch, TensorFlow, Keras, YOLO, Faster R-CNN, MediaPipe, Dlib, Model Evaluation, Transfer Learning

Generative AI & LLMs: OpenAI, Claude, Hugging Face, LangChain, LangGraph, RAG (Retrieval-Augmented Generation), AI Agents, MCP (Model Context Protocol), Tool Calling, Prompt Engineering

Search & Retrieval: Embeddings, Semantic Search, Vector Databases: pinecone, chromaDB, FAISS, Milvus, Qdrant, Hybrid Search, Reranking, FTS

Backend & Deployment: FastAPI, Docker, PostgreSQL, SQLAlchemy, Redis, JWT, REST APIs, Azure DevOps

Cloud & MLOps: Azure Machine Learning, Azure Cognitive Services, Azure Data Factory, Azure AI Services

Work Experience

AVL Motors, Data Scientist — Nov 2025 – Present

- Led a team of 4 engineers in delivering production-grade AI solutions using LLMs, RAG, AI Agents, and MCP-based architectures.
- Built DakshVeda, a multi-agent AI platform using LangChain, LangGraph, and MCP for natural language querying of industrial data (OEE, downtime, production), achieving 85%+ query accuracy and supporting real-time interactions via FastAPI WebSockets.
- Developed an RFQ Procurement Assistant leveraging LangChain, LangGraph, OpenAI, Tavily Search, and historical RFQ retrieval to recommend chiller manufacturers, compliance documents, and technical specifications.
- Designed scalable RAG pipelines using Qdrant, SentenceTransformers, semantic search, and vector embeddings for intelligent document retrieval and recommendation systems.
- Engineered Text-to-SQL workflows and scalable backend services using FastAPI, PostgreSQL, Redis, JWT authentication, and Docker for enterprise AI applications.
- Worked on a POC to automate Excel-based expense reporting using Python (pandas), rebuilding pivot-table logic across multiple report views (month-wise, segment-wise, cost-center-wise, GPN-wise, etc.) to validate and reproduce report summaries without manual Excel calculations, saving 1–2 days of manual effort per report cycle.

Netsmartz, AI/ML Engineer — Feb 2023 – Oct 2025

- Built churn prediction system using Logistic Regression, XGBoost, and LDA with 90%+ accuracy.
- Engineered a facial diagnostic tool for PTSD, TBI, and Depression using YOLO, faster-RCNN and OpenCV. Integrated Claude (Anthropic) via few-shot prompting for low-data features to improve predictions.
- Built enterprise document intelligence and multitenant chatbots using LangChain, RAG, Milvus, semantic search, JWT authentication, and Redis caching.
- Worked on a POC for caregiver scheduling optimization using a Genetic Algorithm, designing a constraint-aware assignment engine that matched caregivers to client visits based on availability, skills, travel distance, and working-hour limits, reducing scheduling conflicts and manual planning effort.

Key Projects

DakshVeda – AI-Powered Industrial Operations Assistant

- Built a multi-agent conversational AI system using LangChain and LangGraph to enable natural language querying of industrial metrics like OEE, downtime, and production.

- Designed hierarchical agent workflows with intent classification, query decomposition, contextual memory, and MCP-based tool calling; exposed enterprise APIs as MCP servers enabling agents to retrieve real-time operational insights.
- Built multilingual and persona-aware conversational experiences, adapting responses based on user roles and language preferences while maintaining contextual accuracy.
- Implemented RAG pipelines using Qdrant and SentenceTransformers for semantic retrieval across operational documents and enterprise knowledge sources.
- Developed real-time FastAPI WebSocket services with JWT authentication, RBAC, Redis caching, and and provider-agnostic OpenAI/Claude integrations, achieving 85%+ accuracy on complex multi-step queries.

RFQ Chiller Field Predictor

- Led a team of 3 engineers in developing a multi-agent AI procurement assistant using LangGraph, OpenAI, and Tavily Search to predict chiller manufacturers, compliance documents, and technical specifications from green-field RFQs.
- Built a hybrid recommendation engine combining Text-to-SQL, historical RFQ analysis, and live web-grounded search to generate procurement recommendations from structured and unstructured data sources.
- Engineered schema-aware Text-to-SQL workflows that retrieve similar historical RFQs from PostgreSQL and provide contextual examples to LLMs for intelligent field prediction and autofill.
- Developed an AI decision layer leveraging tool calling and confidence-based ranking to evaluate multiple recommendations and select the most relevant output for end users.
- Implemented human-in-the-loop review workflows, enabling engineers to approve or override AI-generated recommendations before persistence.

Facial-dx

- Designed a facial diagnostic system to detect PTSD, TBI, and depression by analyzing facial features with model-specific pipelines.
- Implemented facial feature extraction using YOLO (object detection & segmentation), OpenCV, MediaPipe, Dlib, and Faster-RCNN for traits like dark circles, forehead lines, pupil size, mouth tilt, and nose shape; integrated Claude (Anthropic) with few/zero-shot prompting for low-data features.
- Structured outputs using Pydantic, stored predictions in PostgreSQL, and applied doctor-defined rules to generate mental health outcomes with over 80% accuracy.

Caregiver Churn Prediction

- Built a classification pipeline using Logistic Regression, XGBoost, and LDA, achieving over 90% accuracy through feature engineering and hyperparameter tuning.
- Conducted data cleaning, EDA, and applied GridSearchCV; evaluated models using ROC-AUC, F1-score, and confusion matrix. Developed real-time prediction APIs using FastAPI and deployed using Docker.
- Tools: Python, Pandas, scikit-learn, XGBoost, FastAPI, Docker

Chatbot Platform

- Built a document-based Q&A chatbot using IBM Foundation Models to extract answers from unstructured PDFs, DOCX, and TXT files.
- Embedded documents into Milvus for semantic search and vector-based retrieval with contextual relevance.
- Developed a multitenant backend using JWT authentication, Redis caching, email verification, and secure role-based access.

Education

Rungta College of Engineering and Technology, Bhilai	Aug 2019 – June 2023
Bachelor's Degree in Computer Science	**Percentage: 82.5%**
St. Xavier's Sr. Sec. School, Bhilai	April 2018 – June 2019
12th(CBSE Board)	**Percentage: 76.5%**
St. Xavier's Hr. Sec. School, Bhilai	April 2016 – June 2017
10th(CG. Board)	**Percentage: 84.5%**

Certificates and Accomplishments

- **Microsoft Certified: Azure Data Scientist Associate** (link)
- **Team Extra Miler Award** twice for outstanding contributions and collaboration.